



S. J. P. N. TRUST'S

HIRASUGAR INSTITUTE OF TECHNOLOGY, NIDASOSHI

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Department of Computer Science & Engineering

MACHINE LEARNING

Module 5:

Chapter 8: Cluster analysis

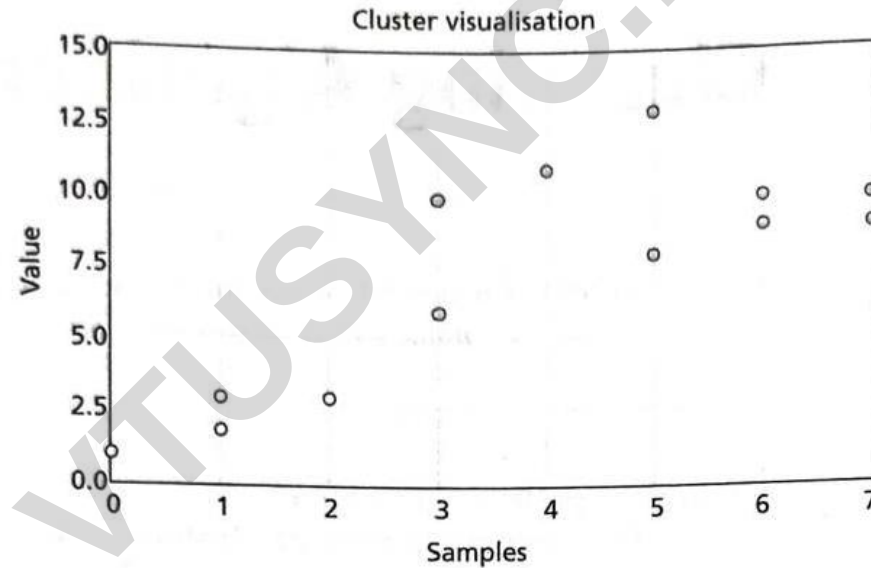
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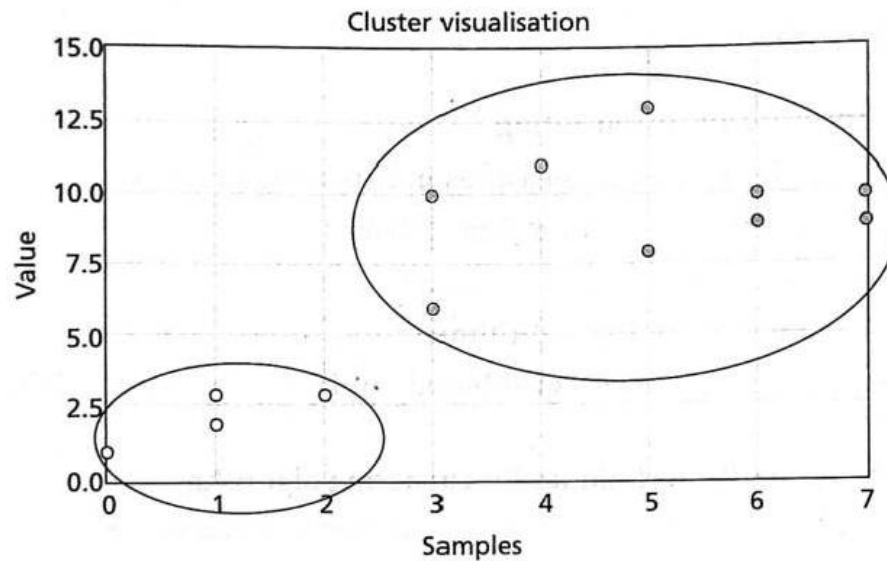
INTRODUCTION TO CLUSTERING APPROACHES

- Cluster analysis is the fundamental task of **unsupervised learning**.
- Cluster analysis is a technique of **partitioning a collection of unlabelled objects, with many attributes, into meaningful disjoint groups or clusters**.
- This is **done using a trial and error approach** as there are no supervisors available as in classification.
- The input for cluster analysis is examples or samples.
- These are known as objects, data points or data instances.

INTRODUCTION TO CLUSTERING APPROACHES



(a)



(b)

Figure 13.1: (a) Data Samples (b) Clusters' Description

Differences between Classification and Clustering

Clustering	Classification
Unsupervised learning and cluster formation are done by trial and error as there is no supervisor	Supervised learning with the presence of a supervisor to provide training and testing data
Unlabelled data	Labelled data
No prior knowledge in clustering	Knowledge of the domain is must to label the samples of the dataset
Cluster results are dynamic	Once a label is assigned, it does not change

Applications of Clustering

- Grouping based on customer buying patterns
- Profiling of customers based on lifestyle
- In information retrieval applications (like retrieval of a document from a collection of documents)
- Identifying the groups of genes that influence a disease
Identification of organs that are similar in physiology functions
- Taxonomy of animals, plants in Biology
- Clustering based on purchasing behaviour and demography
- Document indexing
- Data compression by grouping similar objects and finding duplicate objects

Advantages	Disadvantages
Cluster analysis algorithms can handle missing data and outliers.	Cluster analysis algorithms are sensitive to initialization and order of the input data.
Can help classifiers in labelling the unlabelled data. Semi-supervised algorithms use cluster analysis algorithms to label the unlabelled data and then use classifiers to classify them.	Often, the number of clusters present in the data have to be specified by the user.
It is easy to explain the cluster analysis algorithms and to implement them.	Scaling is a problem.
Clustering is the oldest technique in statistics and it is easy to explain. It is also relatively easy to implement.	Designing a proximity measure for the given data is an issue.

PROXIMITY MEASURES

- Clustering algorithms need a measure to find the similarity or dissimilarity among the objects to group them.
- **Similarity and dissimilarity are collectively known as proximity measures.**
- Often, the distance measures are used to find similarity between two objects, say i and j .
- Distance measures are known as dissimilarity measures, as these indicate how one object is different from another.
- Distance between two objects, say i and j , is denoted by the symbol D_{ij} .

The properties of the distance measures are:

1. D_{ij} is always positive or zero.
2. $D_{ij} = 0$, i.e., the distance between the object to itself is 0.
3. $D_{ij} = D_{ji}$. This property is called symmetry.
4. $D_{ij} \leq D_{ik} + D_{kj}$. This property is called triangular inequality.

Quantitative Variables

- **Euclidean Distance** It is also called as L_2 norm. It can be defined as the square root of squared differences between the coordinates of a pair of objects.
- The Euclidean distance between objects x_i and x_j with k features is given as follows:

$$\text{Distance}(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}$$

- **City Block Distance** City block distance is known as Manhattan distance. This is also known as boxcar, absolute value instance, Manhattan distance, Taxicab or L_1 norm.
- The formula for finding the distance is given as follows:

$$\text{Distance}(x_i, x_j) = \sum_{k=1}^n |x_{ik} - x_{jk}|$$

Quantitative Variables

- **Chebyshev Distance** Chebyshev distance is known as maximum value distance.
- This is the absolute magnitude of the differences between the coordinates of a pair of objects.
- This distance is called supremum distance or L. or L. norm.
- The formula for computing Chebyshev distance is

$$\text{Distance } (x_i, x_j) = \max_k |x_{ik} - x_{jk}|$$

Example 13.1: Suppose, if the coordinates of the objects are (0, 3) and (5, 8), then what is the Chebyshev distance?

Solution: The Euclidean distance using Eq. (13.1) is given as follows:

$$\begin{aligned}\text{Distance } (x_i, x_j) &= \sqrt{(0 - 5)^2 + (3 - 8)^2} \\ &= \sqrt{50} = 7.07\end{aligned}$$

The Manhattan distance using Eq. (13.2) is given as follows:

$$\text{Distance } (x_i, x_j) = |(0 - 5) + (3 - 8)| = 10$$

The Chebyshev distance using Eq. (13.3) is given as follows:

$$\text{Max } \{|0 - 5|, |3 - 8|\} = \text{Max } \{5, 5\} = 5$$

Quantitative Variables

- **Minkowski Distance** In general, all the above distance measures can be generalized as:

$$\text{Distance}(x_i, x_j) = \left(\sum |x_{ik} - x_{jk}|^r \right)^{\frac{1}{r}}$$

- This is called Minkowski distance.
- Here, r is a parameter.
- When the value of r is 1, the distance measure is called city block distance.
- When the value of r is 2, the distance measure is called Euclidean distance.
- When, r is ∞ , then this is Chebyshev distance.

Binary Attributes

- Binary attributes have only two values.
- For finding the distance among objects with binary objects, the contingency Table 13.3 can be used

Table 13.3: Contingency Table

Attributes Matching	0	1
0	a	b
1	c	d

- Let x and y be the objects consisting of N -binary objects. Then, the contingency table can be constructed by counting the number of matching of transitions, 0-0, 0-1, 1-0 and 1-1.
- In other words, 'a' is the number of attributes where x attribute is 0 and y attribute is 0. 'b' is the number of attributes where x attribute is 0 and y attribute is 1, 'c' is the number of attributes where x attribute is 1 and y attribute is 0 and 'd' is the number of attributes where x attribute is 1 and y attribute is 1.

Simple Matching Coefficient (SMC)

- SMC is a simple distance measure and is defined as the ratio of number of matching attributes and the number of attributes.
- The formula is given as:

$$\frac{a + d}{a + b + c + d}$$

- Jaccard Coefficient Jaccard coefficient is another useful measure for and is given as follows:

$$J = \frac{d}{b + c + d}$$

Hamming Distance

- Hamming distance is another useful measure that can be used for **knowing the sequence of characters or binary values.**
- It indicates the number of positions at which the characters or binary bits are different.
- For example, the hamming distance between $x = (1\ 0\ 1)$ and $y = (1\ 1\ 0)$ is 2 as x and y differ in two positions.
- The distance between two words, say wood and hood is 1, as they differ in only one character.

categorical Variables

- In many cases, categorical values are used.
- It is just a code or symbol to represent the values.
- For example, for the attribute Gender, a code 1 can be given to female and 0 can be given to male.
- To calculate the distance between two objects represented by variables, we need to find only whether they are equal or not.
- This is given as:

$$\text{Distance } (x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

Ordinal Variables

- Ordinal variables are like categorical values but with an inherent order. For example, designation is an ordinal variable.
- If job designation is 1 or 2 or 3, it means code 1 is higher than 2 and code 2 is higher than 3. It is ranked as $1 > 2 > 3$.
- Let us assume the designations of office **employees are clerk, supervisor, manager and general manager**. These can be designated as numbers as clerk = 1, supervisor = 2, manager = 3 and general manager = 4. Then, the distance between employee X who is a clerk and Y who is a manager can be obtained as:

$$\text{Distance } (X, Y) = \frac{|\text{position } (X) - \text{position } (Y)|}{n - 1} \quad (13.8)$$

Here, position (X) and position(Y) indicate the designated numerical value. Thus, the distance between X (Clerk = 1) and Y (Manager = 3) using Eq. (13.8) is given as:

$$\text{Distance } (X, Y) = \frac{|\text{position } (X) - \text{position } (Y)|}{n - 1} = \frac{|1 - 3|}{4 - 1} = \frac{2}{3} \approx 0.66$$

HIERARCHICAL CLUSTERING ALGORITHMS

- Hierarchical methods produce a nested partition of objects with hierarchical relationships among objects.
- Often, the hierarchy relationship is shown in the form of a dendrogram.
- Hierarchical methods include categories, agglomerative methods and divisive methods.
- In agglomerative methods, initially all individual samples are considered as a cluster, that is a cluster with a single element.
- Then, they are merged and the process is continued to get a single cluster.
- Agglomerative methods merge clusters to reduce the number of clusters.
- This is repeated each time while merging two closest clusters to get a single cluster.

Algorithm 13.1: Agglomerative Clustering

1. Place each N sample or data instance into a separate cluster. So, initially N clusters are available.
2. Repeat the following steps until a single cluster is formed:
 - (a) Determine two most similar clusters.
 - (b) Merge the two clusters into a single cluster reducing the number of clusters as $N-1$.
3. Choose resultant clusters of step 2 as result.

Single Linkage or MIN Algorithm

- The idea behind this approach is proximity among clusters.
- In single linkage algorithm, the smallest distance (x, y) , where x is from one cluster and y is from another cluster, is the distance between all possible pairs of the two groups or clusters (or simply the smallest distance of two points where points are in different clusters) and is used for merging the clusters.
- This corresponds to finding of minimum spanning tree (MST) of a graph.

Example 13.4: Consider the array of points as shown in the following Table. 13.4.

Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Apply single linkage algorithm.

Solution: The similarity table among the variables is computed as shown in Table 13.4. Euclidean distance is computed as shown in the following Table 13.5.

Table 13.5: Proximity Matrix

Objects	0	1	2	3	4
0	–	4.123	7.211	17.804	27.295
1		–	3.606	14.142	23.324
2			–	10.630	20.124
3				–	10.198
4					–

The minimum distance is 3.606. Therefore, the items 1 and 2 are clustered together. The resultant is shown in Table 13.6.

Table 13.6: After Iteration 1

Clusters	{1, 2}	0	3	4
{1, 2}	–	4.123	10.630	20.124
0		–	17.804	27.295
3			–	10.198
4				–

The distance between the group {1, 2} and items {0}, {3} and {4} is computed using the formula:

$$D_{SL}(C_i, C_j) = \text{minimum}_{a \in C_i, b \in C_j} d(a, b) \quad (13.10)$$

Here, D_{SL} is the single linkage distance, C_i, C_j are clusters and $d(a, b)$ is the distance between the elements a and b .

Thus, the distance between {1, 2} and {0} is:

$$\text{Minimum} \{\{1, 0\}, \{2, 0\}\} = \text{Minimum} \{4.123, 7.211\} = 4.123$$

The distance between {1, 2} and {3} is given as:

$$\text{Minimum} \{\{1, 3\}, \{2, 3\}\} = \text{Minimum} \{14.412, 10.630\} = 10.630$$

The distance between {1, 2} and {4} is given as:

$$\text{Minimum} \{\{1, 4\}, \{2, 4\}\} = \text{Minimum} \{23.324, 20.124\} = 20.124$$

This is shown in Table 13.6. The minimum distance of Table 13.6 is 4.123. Therefore, $\{0, 1, 2\}$ is clustered together. This result is shown in Table 13.7.

Table 13.7: After Iteration 2

Clusters	$\{0, 1, 2\}$	3	4
$\{0, 1, 2\}$	–	10.630	20.124
3		–	10.198
4			–

Thus, the distance between $\{0, 1, 2\}$ and $\{3\}$ using Eq. (13.10) is given as

$$\text{Minimum } \{\{0, 3\}, \{1, 3\}, \{2, 3\}\} = \text{Minimum } \{17.804, 14.142, 10.630\} = 10.630.$$

Thus, the distance between $\{0, 1, 2\}$ and $\{4\}$ is:

$$\text{Minimum } \{\{0, 4\}, \{1, 4\}, \{2, 4\}\} = \text{Minimum } \{27.295, 23.324, 20.124\} = 20.124.$$

This is shown in Table 13.7. The minimum is 10.198.

Therefore, the items {3, 4} are merged. And, the items {1, 2, 3} and {4, 5} are merged. The resultant is shown in Table 13.8.

Table 13.8: After Iteration 3

Clusters	{0, 1, 2}	{3, 4}
{0, 1, 2}	–	?
{3, 4}		–

The computation of ? in Table 13.8 is not needed as no other items are left. Therefore, the clusters {0, 1, 2} and {3, 4} are merged.

Dendrograms are used to plot the hierarchy of clusters. Dendrogram for the above clustering is shown in the following Figure 13.2.

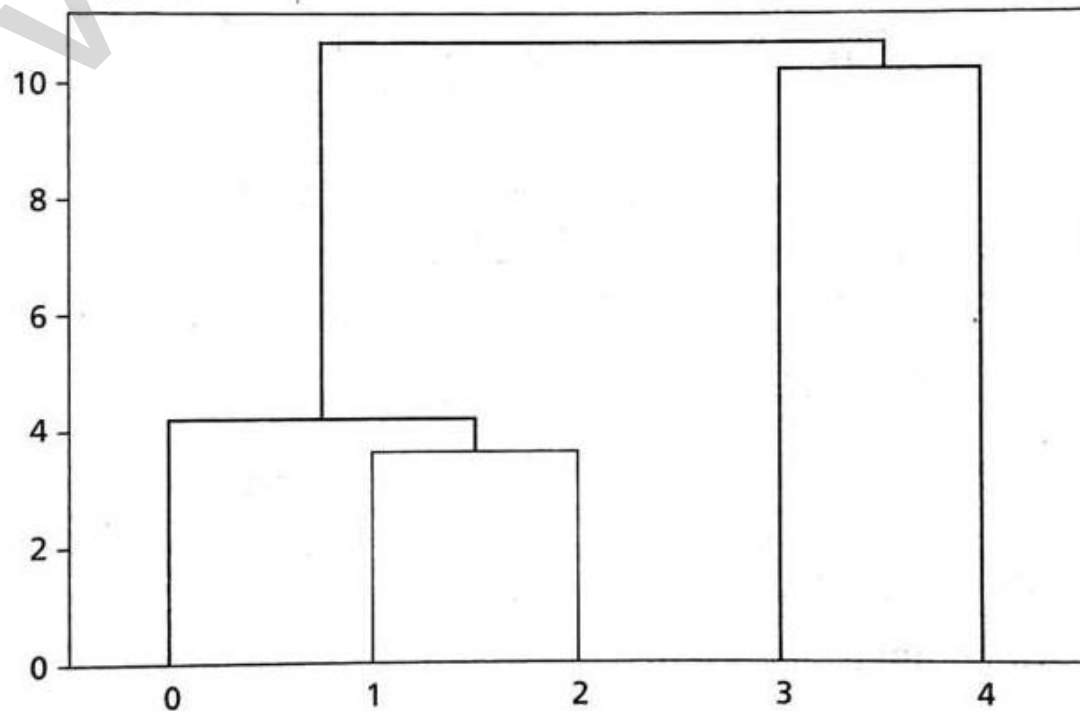


Figure 13.2: Dendrogram for Table 13.4

Complete Linkage or MAX or Clique

- In complete linkage algorithm, **the distance (x, y)**, (where x is from one cluster and y is from another cluster), **is the largest distance between all possible pairs of the two groups or clusters** (or simply the largest distance of two points where points are in different clusters) as given below.
- It is used for merging the clusters.

$$D_{cl}(C_i, C_j) = \text{maximum}_{a \in C_i, b \in C_j} d(a, b)$$

Complete Linkage or MAX or Clique

Dendrogram for the above clustering is shown in Figure 13.3.

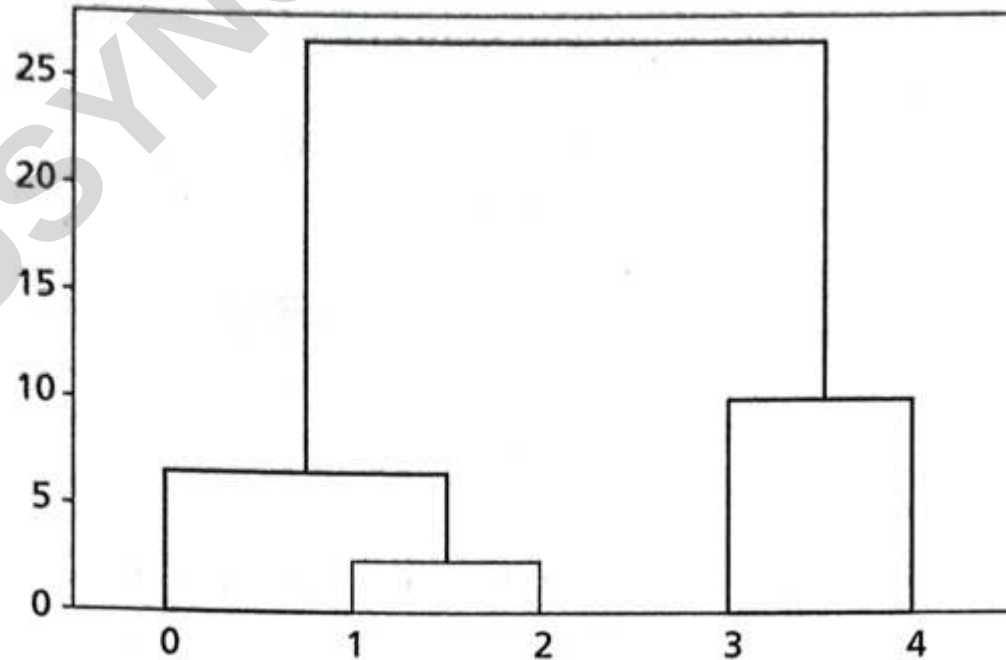


Table 13.4: Sample Data

Objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Figure 13.3: Dendrogram for Table 13.4

Average Linkage

- In case of an average linkage algorithm, **the average distance of all pairs of points across the clusters is used to form clusters.**
- The average value computed between clusters c_i , C_j is given as follows:

$$D_{AL}(C_i, C_j) = \frac{1}{m_i m_j} \sum_{a \in C_i, b \in C_j} d(a, b)$$

Here, m_i and m_j are sizes of the clusters.

The dendrogram for Table 13.4 is given in Figure 13.4.

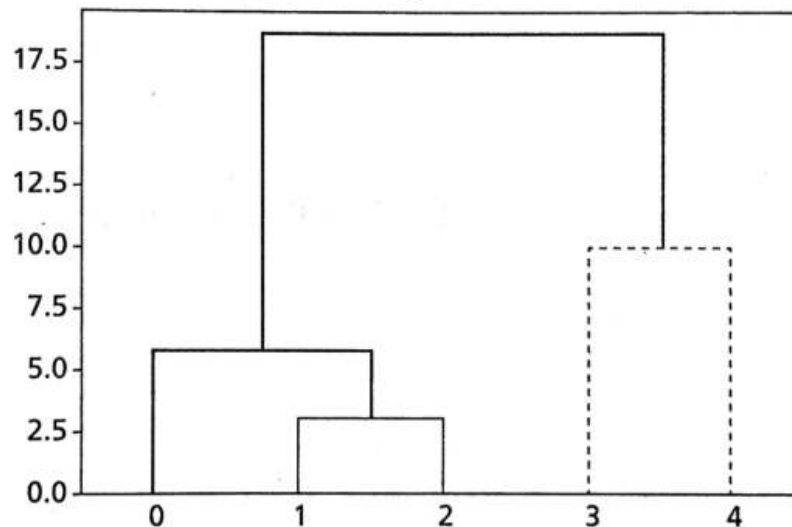


Figure 13.4: Dendrogram for Table 13.4

Mean-Shift Clustering Algorithm

- **Mean-shift** is a non-parametric and hierarchical clustering algorithm.
- This algorithm is also known as mode seeking algorithm **or a sliding window algorithm.**
- It has many applications in image processing and computer vision.
- **The algorithm slowly moves from its initial position towards the dense regions.**
- **The algorithm uses a window, which is basically a weighting function.**
- Gaussian window is a good example of a window.
- The radius of the kernel is called bandwidth.
- The entire window is called a kernel.
- The window is based on the concept of kernel density function and its aim is to find the underlying data distribution.
- The method of calculation of mean is dependent on the choice of windows.
- **If a Gaussian window is chosen, then every point is assigned a weight that decreases as the distance from the kernel center increases.**

Mean-Shift Clustering Algorithm

Algorithm 13.2: Mean-Shift Clustering

Step 1: Design a window.

Step 2: Place the window on a set of data points.

Step 3: Compute the mean for all the points that come under the window.

Step 4: Move the center of the window to the mean computed in step 3. Thus, the window moves towards the dense regions. The movement to the dense region is controlled by a mean shift vector. The mean shift vector is given as:

$$v_s = \frac{1}{K} \sum_{x_i \in S_k} (x_i - x) \quad (13.13)$$

Here, K is the number of points and S_k is the data points where the distance from data points x_i and centroid of the kernel x is within the radius of the sphere. Then, the centroid is updated as $x = x + v_s$.

Step 5: Repeat the steps 3–4 for convergence. Once convergence is achieved, no further points can be accommodated.

Mean-Shift Clustering Algorithm

Advantages

- No model assumptions
- Suitable for all non-convex shapes
- Only one parameter of the window, that is, bandwidth is required
- Robust to noise
- **No** issues of local minima or premature termination

Mean-Shift Clustering Algorithm

Disadvantages

- Selecting the bandwidth is a challenging task. If it is larger, then many clusters are missed.
- If it is small, then many points are missed and convergence occurs as the problem.
- The number of clusters cannot be specified and user has no control over this parameter.

PARTITIONAL CLUSTERING ALGORITHM (K-Means)

- k-means algorithm is a straightforward **iterative partitional algorithm**.
- Here, **k stands for the user specified requested clusters** as users are not aware of the clusters that are present in the dataset.
- The k-means algorithm assumes that the clusters do not overlap. Therefore, a sample or data point can belong to only one cluster in the end.
- Also, this algorithm can detect clusters of shapes like circular or spherical.
- **Assigning each sample or data point to the k cluster centers based on its distance and the centroid of the clusters.**
- This distance should be minimum.
- As a new sample is added, new computation of mean vectors of the points for that cluster to which sample is assigned is required.
- Therefore, this iterative process is continued until no change of instances to clusters is noticed.

Algorithm 13.3: *k*-means Algorithm

- Step 1:** Determine the number of clusters before the algorithm is started. This is called k .
- Step 2:** Choose k instances randomly. These are initial cluster centers.
- Step 3:** Compute the mean of the initial clusters and assign the remaining sample to the closest cluster based on Euclidean distance or any other distance measure between the instances and the centroid of the clusters.
- Step 4:** Compute new centroid again considering the newly added samples.
- Step 5:** Perform the steps 3–4 till the algorithm becomes stable with no more changes in assignment of instances and clusters.

k -means can also be viewed as greedy algorithm as it involves **partitioning n samples to k clusters to minimize Sum of Squared Error (SSE)**. SSE is a metric that is a measure of error that gives the sum of the squared Euclidean distances of each data to its closest centroid. It is given as:

given as:

$$SSE = \sum_{i=1}^k \text{dist}(c_i, x)^2 \quad (13.14)$$

Here, c_i is the centroid of the i^{th} cluster, x is the sample or data point and dist is the Euclidean distance. The aim of the k -means algorithm is to minimize SSE.

K-Means Clustering

- **Advantages**
 - Simple
 - Easy to implement
- **Disadvantages**
 - ***It*** is sensitive to initialization process as change of initial points leads to different clusters.
 - If the samples are large, then the algorithm takes a lot of time.
- **Now to Choose the Value of k ?**
- It is obvious that k is the user specified value specifying the number of clusters that are present.
- Obviously, there are no standard rules available to pick the value of k .
- Normally, the k-means algorithm is run with multiple values of k and within group variance (sum of squares of samples with its centroid) and plotted as a line graph. This plot is called Elbow curve. The optimal or best value of k can be determined from the graph.
- The optimal value of k is identified by the flat or horizontal part of the Elbow curve.

Example 13.5: Consider the following set of data given in Table 13.9. Cluster it using k -means algorithm with the initial value of objects 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds.

Table 13.9: Sample Data

Objects	X-coordinate	Y-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

Solution: As per the problem, choose the objects 2 and 5 with the coordinate values. Hereafter, the objects' id is not important. The samples or data points (4, 6) and (12, 4) are started as two clusters as shown in Table 13.10.

Initially, centroid and data points are same as only one sample is involved.

Table 13.10: Initial Cluster Table

Cluster 1	Cluster 2
(4, 6)	(12, 4)
Centroid 1 (4, 6)	Centroid 2 (12, 4)

Iteration 1: Compare all the data points or samples with the centroid and assign to the nearest sample. Take the sample object 1 (2, 4) from Table 13.9 and compare with the centroid of the clusters in Table 13.10. The distance is 0. Therefore, it remains in the same cluster. Similarly, consider the remaining samples. For the object 1 (2, 4), the Euclidean distance between it and the centroid is given as:

$$\text{Dist (1, centroid 1)} = \sqrt{(2 - 4)^2 + (4 - 6)^2} = \sqrt{8}$$

$$\text{Dist (1, centroid 2)} = \sqrt{(2 - 12)^2 + (4 - 4)^2} = \sqrt{100} = 10$$

Object 1 is closer to the centroid of cluster 1 and hence assign it to cluster 1. This is shown in Table 13.11. Object 2 is taken as centroid point.

For the object 3 (6, 8), the Euclidean distance between it and the centroid points is given as:

$$\text{Dist (3, centroid 1)} = \sqrt{(6 - 4)^2 + (8 - 6)^2} = \sqrt{8}$$

$$\text{Dist (3, centroid 2)} = \sqrt{(6 - 12)^2 + (8 - 4)^2} = \sqrt{52}$$

Object 3 is closer to the centroid of cluster 1 and hence remains in the same cluster 1.

Proceed with the next point object 4(10, 4) and again compare it with the centroids in Table 13.10.

$$\text{Dist (4, centroid 1)} = \sqrt{(10 - 4)^2 + (4 - 6)^2} = \sqrt{40}$$

$$\text{Dist (4, centroid 2)} = \sqrt{(10 - 12)^2 + (4 - 4)^2} = \sqrt{4} = 2$$

Object 4 is closer to the centroid of cluster 2 and hence assign it to the cluster table. Object 4 is in the same cluster. The final cluster table is shown in Table 13.11.

Obviously, Object 5 is in Cluster 3. Recompute the new centroids of cluster 1 and cluster 2. They are (4, 6) and (11, 4), respectively.

Table 13.11: Cluster Table After Iteration 1

Cluster 1	Cluster 2
(4, 6)	(10, 4)
(2, 4)	(12, 4)
(6, 8)	
Centroid 1 (4, 6)	Centroid 2 (11, 4)

The second iteration is started again with the Table 13.11.

Obviously, the point (4, 6) remains in cluster 1, as the distance of it with itself is 0. The remaining objects can be checked. Take the sample object 1 (2, 4) and compare with the centroid of the clusters in Table 13.12.

$$\text{Dist (1, centroid 1)} = \sqrt{(2 - 4)^2 + (4 - 6)^2} = \sqrt{8}$$

$$\text{Dist (1, centroid 2)} = \sqrt{(2 - 11)^2 + (4 - 4)^2} = \sqrt{81} = 9$$

Object 1 is closer to centroid of cluster 1 and hence remains in the same cluster. Take the sample object 3 (6, 8) and compare with the centroid values of clusters 1 (4, 6) and cluster 2 (11, 4) of the Table 13.12.

$$\text{Dist (3, centroid 1)} = \sqrt{(6 - 4)^2 + (8 - 6)^2} = \sqrt{8}$$

$$\text{Dist (3, centroid 2)} = \sqrt{(6 - 11)^2 + (8 - 4)^2} = \sqrt{41}$$

Object 3 is closer to centroid of cluster 1 and hence remains in the same cluster. Take the sample object 4 (10, 4) and compare with the centroid values of clusters 1 (4, 6) and cluster 2 (11, 4) of the Table 13.12:

$$\text{Dist (4, centroid 1)} = \sqrt{(10 - 4)^2 + (4 - 6)^2} = \sqrt{40}$$

$$\text{Dist (3, centroid 2)} = \sqrt{(10 - 11)^2 + (4 - 4)^2} = \sqrt{1} = 1$$

Object 3 is closer to centroid of cluster 2 and hence remains in the same cluster. Obviously, the sample (12, 4) is closer to its centroid as shown below:

$$\text{Dist (5, centroid 1)} = \sqrt{(12 - 4)^2 + (4 - 6)^2} = \sqrt{68}$$

$\text{Dist (5, centroid 2)} = \sqrt{(12 - 11)^2 + (4 - 4)^2} = \sqrt{1} = 1$. Therefore, it remains in the same cluster. Object 5 is taken as centroid point.

The final cluster Table 13.12 is given below:

Table 13.12: Cluster Table After Iteration 2

Cluster 1	Cluster 2
(4, 6)	(10, 4)
(2, 4)	(12, 4)
(6, 8)	
Centroid (4, 6)	Centroid (11, 4)

There is no change in the cluster Table 13.12. It is exactly the same; therefore, the *k*-means algorithm terminates with two clusters with data points as shown in the Table 13.12.

DENSITY-BASED METHODS

- Density-based spatial clustering of applications with noise (DBSCAN) is one of the density-based algorithms. Density of a region represents the region where many points above the specified threshold are present.
- In a density-based approach, the clusters are regarded as dense regions of objects that are separated by regions of low density such as noise. This is same as a human's intuitive way of observing clusters.
- The concept of density and connectivity is based on the local distance of neighbours.
- The functioning of this algorithm is based on two parameters, the size of the neighbourhood (ϵ) and the minimum number of points (m).
- Core point — A point is called a core point if it has more than specified number of points (m) within ϵ -neighbourhood.
- Border point — A point is called a border point if it has fewer than ' m ' points but is a neighbour of a core point.
- Noise point — A point that is neither a core point nor border point.

DENSITY-BASED METHODS

- The main idea is that every data point or sample should have at least a minimum number of neighbours in a neighbourhood.
 - The neighbourhood of radius ϵ should have at least m points.
 - The notion of density connectedness determines the quality of the algorithm.
 - The following connectedness measures are used for this algorithm.
1. Direct density reachable — The point X is directly reachable from Y , if: X is in the neighbourhood of Y and Y is a core point
 2. Densely reachable — The point X is densely reachable from Y , if there is a set of core points that leads from Y to X .
 3. Density connected — X and Y are densely connected if Z is a core point and thus points X and Y are densely reachable from Z .

DENSITY-BASED METHODS

Algorithm 13.4: DBSCAN

- Step 1:** Randomly select a point p . Compute distance between p and all other points.
- Step 2:** Find all points from p with respect to its neighbourhood and check whether it has minimum number of points m . If so, it is marked as a core point.
- Step 3:** If it is a core point, then a new cluster is formed, or existing cluster is enlarged.
- Step 4:** If it is a border point, then the algorithm moves to the next point and marks it as visited.
- Step 5:** If it is a noise point, they are removed.
- Step 6:** Merge the clusters if it is mergeable, $\text{dist}(c_i, c_j) < \epsilon$.
- Step 7:** Repeat the process 3–6 till all points are processed.

- **Advantages**

1. No need for specifying the number of clusters beforehand
2. The algorithm can detect clusters of any shapes
3. Robust to noise
4. Few parameters are needed

GRID-BASED APPROACH

- Grid-based approach is a space-based approach. It partitions space into cells, the given data is fitted on the cells for cluster formation.
- There are three important concepts that need to be mastered for understanding the grid-based schemes. They are:
 1. Subspace clustering
 2. Concept of dense cells
 3. Monotonicity property

Subspace Clustering

- Grid-based algorithms are useful for clustering high-dimensional data, that is data with many attributes.
- Some data like gene data may have millions of attributes. Every attribute is called a dimension. But all the attributes are not needed, as in many applications one may not require all the attributes.
- For example, an employee's address may not be required for profiling his diseases.
- Age may be required in that case. So, one can conclude that only a subset of features is required.
- For example, one may be interested in grouping gene data with similar characteristics or organs that have similar functions.
- Finding subspaces is difficult. For example, N dimensions may have $2^N - 1$ subspaces. Exploring all the subspaces is a difficult task. Here, only the CLIQUE algorithms are useful for exploring the subspaces.
- CLIQUE (Clustering in Quest) is a grid-based method for finding clustering in subspaces.
- CLIQUE uses a multiresolution grid data structure.

Concept of Dense Cells

- CLIQUE partitions each dimension into several overlapping intervals and intervals it into cells.
- Then, the algorithm determines whether the cell is dense or sparse. The cell is considered dense if it exceeds a threshold value, say T . Density is defined as the ratio of number of points and volume of the region.
- In one pass, the algorithm finds the number of cells, number of points, etc. and then combines the dense cells.
- For that, the algorithm uses the contiguous intervals and a set of dense cells.

Algorithm 13.5: Dense Cells

Step 1: Define a set of grid points and assign the given data points on the grid.

Step 2: Determine the dense and sparse cells. If the number of points in a cell exceeds the threshold value τ , the cell is categorized as dense cell. Sparse cells are removed from the list.

Step 3: Merge the dense cells if they are adjacent.

Step 4: Form a list of grid cells for every subspace as output.

Monotonicity Property

- CLIQUE uses anti-monotonicity property or apriori property of the famous apriori algorithm.
- It means that all the subsets of a frequent item should be frequent. Similarly, if the subset is infrequent, then all its supersets are infrequent as well. Based on the apriori property, one can conclude that a k -dimensional cell has r points if and only if every $(k - 1)$ dimensional projections of this cell have at least r points. So like association rule mining that uses apriori rule, the candidate dense cells are generated for higher dimensions. The

Algorithm 13.6: CLIQUE

Stage 1:

Step 1: Identify the dense cells.

Step 2: Merge dense cells c_1 and c_2 if they share the same interval.

Monotonicity Property

Algorithm 13.6: CLIQUE

Stage 1:

Step 1: Identify the dense cells.

Step 2: Merge dense cells c_1 and c_2 if they share the same interval.

Step 3: Generate Apriori rule to generate $(k + 1)^{\text{th}}$ cell for higher dimension. Then, check whether the number of points cross the threshold. This is repeated till there are no dense cells or new generation of dense cells.

Stage 2:

Step 1: Merging of dense cells into a cluster is carried out in each subspace using maximal regions to cover dense cells. The maximal region is an hyperrectangle where all cells fall into.

Step 2: Maximal region tries to cover all dense cells to form clusters.

In stage two, CLIQUE starts from dimension 2 and starts merging. This process is continued till the n -dimension.

Monotonicity Property

- Advantages of CLIQUE
 - Insensitive to input order of objects
 - No assumptions of underlying data distributions
 - Finds subspace of higher dimensions such that high-density clusters exist in those subspaces
- Disadvantage
 - The disadvantage of CLIQUE is that tuning of grid parameters, such as grid Size, and finding
 - optimal threshold for finding whether the cell is dense or not is a challenge.